BITCOIN PRICE PREDICTION: A MACHINE LEARNING SAMPLE DIMENSION APPROACH

ABSTRACT

The purpose of the paper is to predict Bitcoin prices using various machine learning techniques. Due to its high volatility attribute, accurate price prediction is the need of the hour for sound investment decision-making. At the offset, this study categorizes Bitcoin price by daily and high-frequency price (5-minute interval price). For its daily and 5-minute interval price prediction, a set of high-dimensional features and fundamental trading features are employed, respectively. Thereafter, we find that statistical methods like Logistic Regression predict daily price with 64.84% accuracy while complex machine learning algorithms like XGBoost predict 5-minute interval price with an accuracy level of 59.4%. This work on Bitcoin price prediction recognizes the significance of sample dimensions in machine learning algorithms.

***Keywords*:** Bitcoin; high-frequency; features; logistic regression; XGBoost

1. INTRODUCTION

Bitcoin was introduced in 2008 by an unknown group of individuals using the name Satoshi Nakamoto and officially came into play in 2009 when its ASCII (American Standard Code for Information Interchange) text file was released as an open-source network. It was fabricated to solve the innate weakness of trustworthy transaction models and was initially elucidated as merely a peer-to-peer digital cash system. Today, it is the most valued cryptocurrency, trading in over 40 exchanges worldwide and undertaking 30 different currencies. As a currency, Bitcoin provides a novel opportunity for accurate price prediction due to its high volatility and relatively young age (McNally et al., 2018). Investors consider Bitcoin to be a kind of speculative investment much like web stocks. Several years of trading and increasing popularity of Bitcoin grabbed attention from across society, especially policymakers, and soon Bitcoin's capitalisation touched its peak at 300 billion US dollars in 2017, almost resembling that of Amazon in 2016 (Yermack, 2015). The University of Cambridge 2017 research estimates show that there have been 2.9 to 5.8 million discrete users employing a cryptocurrency wallet, most of them using bitcoin.

The value of Bitcoin reflects investors' confidence in cryptocurrency (Mai et al., 2018). Bitcoin price fluctuations, due to its idiosyncratic volatility, have plagued investors since it began trading. It is quintessential to find ways to accurately forecast Bitcoin price changes. Previous studies on Bitcoin have predicted its price using two approaches: empirical analysis and analysis of machine learning algorithms. Although existing work has best-utilized machine learning methods for better and precise Bitcoin price prediction, only a few have focused on the feasibility of applying different modelling techniques to samples with varying dimensional features (Chen et al., 2020). This paper aims to predict Bitcoin price by using daily data and accessible high-frequency data, similar to stock market prediction (Madan et al., 2015).

Since seasonality is absent in Bitcoin, machine learning models can be highly effective and useful in its price prediction. However, the selection of appropriate machine learning algorithms depends on the frequency and structure of the data, or else the complexity of such models could lead to overfitting. Previous studies have implemented well-known machine learning algorithms such as recurrent neural network (RNN), long short-term memory (LSTM), support vector machines (SVM), and random forest models. But they simply input data into models without differentiating between data frequency and sample size. Most research in this field concentrates only on accuracy when using machine learning algorithms while ignoring the sample dimension (Chen et al., 2020, McNally et al., 2018).

This study strives to leverage suitable machine learning algorithms to sample dimensions for predicting Bitcoin prices. Motivated by Occam's razor theory (Gamberger & Lavrac, 1997) and making use of our daily and accessible high-frequency datasets, this paper carries out the following methodology. Firstly, we utilize two datasets, one with daily intervals and the other with 5-minute intervals. Going forward, we perform feature engineerings such as high-dimension features for low-frequency data to steer clear of overfitting (Wind, 2014) and low-dimension features for high-frequency data (Rechenthin, 2014). Subsequently, we apply conventional statistical models like Logistic Regression, Linear Discriminant Analysis along with machine learning models (Random Forest, Support Vector Machine, XGBoost, Decision Tree, Quadratic Discriminant Analysis, and k-Nearest Neighbors) for comparing results.

This paper proceeds as follows. We look at the literature review in Section 2, where we compare our subject matter with earlier studies. We elaborate on the data and methodology in Section 3, where we talk about the problem statement, source of data and feature engineering. In Section 4, we discuss the machine learning algorithms and appropriate evaluation indicators, and subsequently, in Section 5 exhibit our empirical results. Section 6 concludes the paper.

2. LITERATURE REVIEW

With regards to Bitcoin price prediction, our literature comprises mostly empirical evidence to analyse its determining factors. The inter-relationship between Bitcoin and search volume of Wikipedia and Google Trends was postulated by Kristoufek (2013). His outcomes reflected a distinct asymmetry between the two, suggesting that speculation and trend patterns drive the valuation of Bitcoin within the cryptocurrency market. Applying time-series analytical techniques to daily Bitcoin data, Ciaian et al. (2016) performed an empirical study on the value factors of Bitcoin within a Barro model (1979) framework. Their results revealed that the economic procedure and attractiveness to investors influence Bitcoin price severely. Bukovina (2016) found that the expository power of sentiment exhibits a positive relationship with the volatility of Bitcoin.

Concrete research on Bitcoin price prediction using machine learning methods is insufficient. Leveraging binary classification to predict Bitcoin price fluctuations was explained by Shah et al. (2014) using a latent source Bayesian regression model. The relationship between Bitcoin price and factors such as technology, sentiment, and economic variables, was demonstrated by Georgoula et al. (2015) utilizing a support vector machine algorithm. In order to analyse the impact of network features on Bitcoin price, Greaves et al. (2015) applied machine learning techniques and obtained

a 55% accuracy level. Analogously, Bitcoin price prediction with an accuracy of 98.7% for daily price and approximately 50-55% for the high-frequency price was achieved via machine learning model implementation by Madan et al. (2015). Additionally, a comparison between the accuracy of recurrent neural network (RNN), long-short term memory (LSTM) and autoregressive integrated moving average (ARIMA) models for Bitcoin price forecast was conducted by McNally et al. (2018) which proved that LSTM achieved the best accuracy level of 52%.

The Occam's razor principle advocates that all other criteria are equal, leveraging machine learning is possible with the simplest of models (Gamberger & Lavrac, 1997). Empirical analysis has substantiated that the overfitting problem is caused by the number of models used and not by their complexity (Zahálka & Železný, 2011). There is considerable debate over the issue of whether simpler models or traditional statistical methods should be applied to predict Bitcoin prices meticulously. Practically, high Vapnik–Chervonenkis dimension models qualify as the simplest solution to most prediction related problems (Tong et al., 2017). In such a scenario, there are two usual paradigms: statistical techniques with massive features and complex models with few features. Stimulated by Occam's razor principle, this paper makes use of simple statistical models to predict Bitcoin's daily price with high-dimensional features and more intricate machine learning models to predict Bitcoin's 5-minute interval price (Chen et al., 2020).

3. DATA AND METHODOLOGY

3.1. Problem statement

The objective of this paper is to forecast Bitcoin prices using machine-learning algorithms. This paper initially performs dimension and feature engineering on Bitcoin daily price and 5-minute interval price datasets. Thereafter, we develop a binary classification algorithm to anticipate the change in signs of Bitcoin prices.

3.2. Data

This study utilizes two datasets. The first includes Bitcoin daily price sourced from the Blockchain.com website. It also comprises of daily data of block size, hash rate, mining difficulty, number of transactions, confirmed transactions per day, mempool transaction count, mempool size, total transaction fees, market capitalization, estimated transaction value, the time between blocks, trades per minute, taken from Bitcoinity.org and Blockchain.com as well as Google trend search volume index from Google.com, and gold spot price from GoldHub, for the period starting from January 1, 2017, to December 31, 2019.

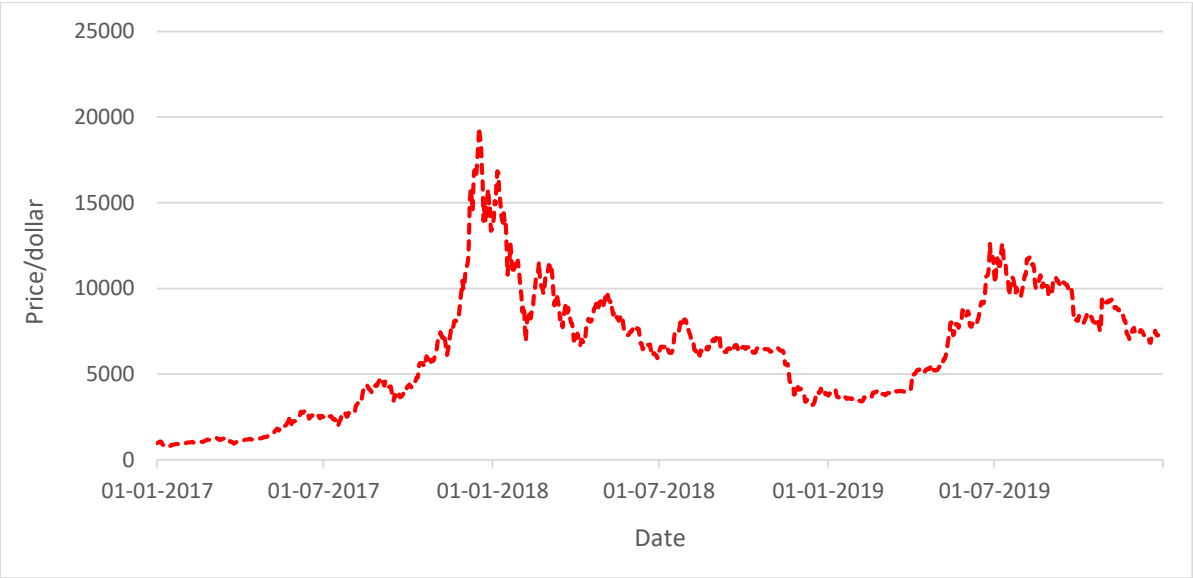

Figure 3.1. Bitcoin Daily Price Distribution

Figure 3.1. shows the distribution of Bitcoin daily price for the period starting January 1, 2017, to December 31, 2019.

Figure 3.2. Bitcoin 5-Minute Interval Price Distribution

Figure 3.2. shows the distribution of Bitcoin 5 – minute interval price for the period November 2, 2016 to June 11, 2018.

From Figure 3.1. we clearly observe that the price kept rising from January 2017 and crashed between January 2018 and February 2019. The succeeding dataset, displayed in Figure 3.2. comprises of 5-minute interval Bitcoin trading prices pulled from the btc-statistics for the period starting from November 2, 2016, to June 11, 2018. We collect the tick data by building an automated web scraper using Python. Since feature data for tick trading were indistinct or unobtainable for very small intervals, in this paper we take into consideration the original features acquired from the source. The feature data includes price, trading volume, open, close, high, and low points for modelling.

Figure 3.2. exhibits that the price kept increasing rapidly from January 2017 till it reached a peak at the beginning of 2018.

3.3. Feature Engineering

Based on the significance of our problem, knowledge domain and understanding, we select the following features for daily price prediction: block size, hash rate, number of transactions, mining difficulty (Hayes, 2015), confirmed transactions per day, mempool transaction count, mempool size, market capitalization, total transaction fees, estimated transaction value, the time between blocks, trades per minute (Madan et al., 2015), Google trend search volume index (Kristoufek, 2013), and gold spot price (Dyhrberg, 2016). Table 3.1. describes the features for daily price prediction included in the model.

Table 3.1. Definition of Daily Data Features

Feature	Definition
Market Capitalisation	The total market value of Bitcoin in the US dollar.
Block Size	A recent chain of bitcoin transactions
Hash Rate	The speed at which a compute is completing an operation within the Bitcoin code
Mining difficulty	A relative difficulty measure to find a new block
Time between blocks	The time interval between one block to other
Trades per minute	The number of Bitcoin traded in minutes
Number of transactions	The number of transactions per day
Confirmed transactions per Day	The number of confirmed transactions per Day
Mempool transaction count	The total number of unconfirmed transactions
Estimated transaction value	The total estimated value of transactions on the Bitcoin blockchain
Total transaction fees	All transaction fees paid to miners
Mempool size	The aggregate size of transactions waiting to be confirmed
Google trend search volume index	The normalized search volume for the inquiry ''Bitcoin'' per day
Gold spot price	XAU gold spot price in US dollars

This paper checks whether the features mentioned should be included in the daily model using the Wrapper Methods (Kohavi & John, 1997). It consists of the forward selection and the backward elimination. Forward selection is an iterative method that starts with having no feature in the model. In each iteration, it keeps adding features

to best improve the model till the addition of a new variable does not improve its performance. The backward elimination process begins with all the features and removes the least significant feature at each iteration to improve the performance of the model. This is repeated until no improvement is observed when features are removed.

Our study leverages the best result for the final features selection and picks the following 12 features: block size, hash rate, mining difficulty, number of transactions, confirmed transactions per day, mempool size, mempool transaction count, market capitalization, estimated transaction value, total transaction fees, Google trend search volume index, and gold spot price. The summary statistics of these features for daily price prediction is provided in Table 3.2.

Table 3.2. Summary Statistics of Features Used for Bitcoin Daily Data Prediction

Feature	Mean	Std.	Min	Max
Market Cap (in billions)	108	60.7	12.2	323
Mining difficulty (in billions)	4940	3860	318	13700
Block size	814600	144075	361627	998175
Number of transactions	278770	61783	131875	490644
Hash rate (in trillions)	3070000	2380000	24100	13000000
Mempool count	17914	28907	3	179540
Mempool size (in millions)	20.4	31.2	0.001	137
Estimated Transaction	182940	94433	38328	632663
Transaction fee	134	172	11	1496
Confirmed Transaction	279093	60517	135274	439549
Price	6301	3504	787	19389
Gold price	1306	89	1146	1546
Google trend	18	23	2	288

Note*: Count is 1095 for all the features.

For the case of Bitcoin 5-minute interval price prediction model, features for tick trading data is unavailable for very small intervals. Therefore, this paper considers the original features, which include open, close, high, low, price and volume.

4. IMPLEMENTATION

In this paper, we develop a binary classification algorithm to envision the sign change of the Bitcoin price, which facilitates traders to make decisions and follow. Binary

dependent variable is denoted as $y \in \{-1,1\}$. Here $y = -1$ which indicates that the Bitcoin price declines, while $y = 1$ specifies that the Bitcoin price increases.

4.1. Machine Learning Algorithms

Our problem statement in this paper is a classification type of problem. Therefore, all the machine learning models used in the work are classification models. We try to select the best classification model(s) for bitcoin price (daily price and 5-minute interval prices) prediction.

This part elaborates on the various machine learning models implemented. Bitcoin daily price and 5-minute interval prices are predicted using machine learning models such as Logistic Regression, Linear Discriminant Analysis, Random Forest, XGBoost, Quadratic Discriminant Analysis, Support Vector Machine, Decision Tree, and k-Nearest Neighbors. We discuss these models briefly and talk about their importance. Although we would like, it is impossible to present the mathematical constructs behind these models to maintain the brevity and to justify the scope of this work. Unlike linear regression, most of these machine learning models are difficult to be expressed in a simple equation as there are too many variables (parameters) and there's no benefit to converting the models into equations. For the same reason, the most application-oriented paper uses selective mathematical formulas, graphical, and structural presentation.

Looking at the results from different models, we find that Logistic Regression and Linear Discriminant Analysis models perform better for daily data with high-dimensional features, while the XGBoost model outperforms all other machine learning algorithms in the case of high-frequency data.

4.1.1. Logistic Regression

Logistic Regression is a simple machine learning technique employed for classification problems. It is a predictive analysis tool that corroborates the concept of probability. It uses an intricate cost function which is defined as the 'Sigmoid function', also called the 'logistic function' rather than a linear function.

4.1.2. Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is a method for dimensionality reduction. It is implemented as a pre-processing step in machine learning and programs of pattern classification. The objective of LDA is to reflect the features in a higher dimensional area onto a lower-dimensional space to avoid overfitting ("curse of dimensionality")

as well as reduce the number of features and computational costs. LDA is a supervised classification approach.

4.1.3. Random Forest

Random Forest is another integral machine learning algorithm. It can be utilized for both classification and regression problems. It follows the concept of ensemble learning, which is a process of mixing multiple classifiers to solve a problem and boost the performance of the model.

Random Forest is a classifier that contains a variety of decision trees on various subsets of the given dataset and takes their average to boost the predictive accuracy of that dataset rather than depending on one decision tree. It takes into account the prediction from each decision tree and on the basis of the bulk votes of predictions, it generates the ultimate output.

4.1.4. XGBoost

XGBoost is also known as Extreme Gradient Boosting. It builds decision trees iteratively. Each tree with a high bias is called a "weak learner". The gradient boosting technique starts by creating a primary simple tree that has an unsatisfactory performance by itself, then it builds another tree which is trained to forecast what the primary tree was not able to and is itself a weak learner too. This procedure goes on by sequentially building more weak learners, all correcting the previous trees until a stopping condition is reached, like the number of trees to create.

4.1.5. Quadratic Discriminant Analysis

Quadratic discriminate analysis (QDA) is a medium for distance discrimination in supervised classification problems. It is assumed that measurements from each class are normally distributed. QDA is much like LDA, except for the fact that it allows for quadratic decision boundaries between classes. The parameters are estimated from training points with maximum likelihood estimation for each class.

4.1.6. Support Vector Machine

Support Vector Machine (SVM) is a supervised machine learning model. It can be used for classification as well as regression problems. It can solve linear and non-linear problems. It uses a technique called the kernel trick to transform data and subsequently find an optimal boundary between the possible outputs. In SVM, the algorithm creates a line or a hyperplane which separates the information into classes.

4.1.7. Decision Tree

A Decision Tree is a machine learning algorithm that is part of a supervised learning technique. It can be employed for regression as well as classification problems. It is a tree-structured classifier. There are two nodes in a decision tree: decision node and leaf node. The decision node is used to make decisions and has multiple branches, whereas a leaf node displays the output of those decisions.

4.1.8. K-Nearest Neighbors

The K-nearest neighbors (KNN) algorithm is a machine learning tool that is a part of the supervised learning technique. It can be used to solve both classification and regression problems. K-nearest neighbors algorithm assumes that similar things lie next to each other.

4.2. Evaluation Indicator

In this research, evaluation indicators are used to assess the performance of machine learning models. One of them is the confusion matrix which is a tabular representation of the performance of machine learning models as well as computes their accuracy in a classification framework.

Table 4.1. Confusion Matrix

		True Class	
		Positive	Negative
Predicted Class	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Table 4.1. indicates that if both actual and predicted are No, then the result is a True Negative. If actual is No and predicted is Yes, then the result is a False Positive. If actual is Yes and predicted is No, then the result is a False Negative. If both actual and predicted are Yes, then the result is a True Positive. The formula of accuracy, precision, recall and F1-score based on a confusion matrix are as follow:

$$\text{Accuracy} = (TP + FN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$\text{F1-score} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$$

Where TP stands for True Positive, TN for True Negative, FP for False Positive, and FN for False Negative.

Accuracy is one of the most intuitive performance measures and is defined as the ratio of correctly predicted observations to the total number of observations. The higher the accuracy, the better is the model. **Precision** is another metric to evaluate the performance of machine learning algorithms. It represents the ratio between correctly predicted positive values to the total predicted positive values. **Recall** is known as sensitivity or true positive rate. It is defined as the ratio of correctly predicted positive observations to the total number of observations. **F1 score** is simply the weighted average of precision and recall. Hence, it takes into consideration both false positives and false negatives simultaneously.

5. RESULTS

Table 5.1. Performance of Different Machine Learning Models on Bitcoin Daily Price

Models	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.648	0.61	0.65	0.63
Linear Discriminant Analysis	0.598	0.66	0.25	0.37
Random Forest	0.529	0.49	0.49	0.49
XGBoost	0.474	0.46	0.89	0.61
Quadratic Discriminant Analysis	0.447	0.44	0.78	0.56
K-Nearest Neighbors	0.47	0.46	0.91	0.61
Decision Tree	0.529	0.48	0.37	0.42
Support Vector Machine	0.461	0.46	1.00	0.63

Table 5.1. illustrates the performance of different machine learning models on Bitcoin's daily price. Appendix 7.1 presents the confusion matrix and classification report tables on bitcoin daily price using different machine learning models. From Table 5.1. we make the following observations. Logistic Regression and Linear Discriminant Analysis fare better than the other machine learning techniques. Logistic Regression gives the best performance with an accuracy of 64.8%, whereas Quadratic

Discriminant Analysis shows an accuracy of a mere 44.7%. Logistic Regression and Linear Discriminant Analysis outperform the other machine learning models, pointing to the fact that prudently selected high-dimensional feature sets to compensate for the simplicity of models in Bitcoin's daily price prediction. Therefore, it can be suggested that for daily bitcoin price prediction, precision can be achieved with simplified models if we have multiple features in the data set.

Table 5.2. Performance of Different Machine Learning Models on Bitcoin 5-Minutes Interval Price

Models	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.593	0.68	0.05	0.11
Linear Discriminant Analysis	0.593	0.71	0.05	0.09
Random Forest	0.542	0.45	0.43	0.44
XGBoost	0.594	0.55	0.16	0.27
Quadratic Discriminant Analysis	0.511	0.42	0.47	0.44
K-Nearest Neighbors	0.54	0.44	0.36	0.39
Decision Tree	0.53	0.43	0.41	0.42
Support Vector Machine	0.56	0.4	0.07	0.12

Table 5.2. exhibits the performance of different machine learning models on Bitcoin's 5-minutes interval price. The confusion matrix and classification report tables on bitcoin's 5-minutes interval price using various machine learning processes are depicted in Appendix 7.2. From our findings in Table 5.2. we evidently observe that the XGBoost model performs the best with an accuracy of 59.4%, while the Decision Tree model displays accuracy of only 53%.

6. CONCLUSION

This study endeavours to anticipate Bitcoin price at different frequencies using different machine learning techniques. Earlier studies simply leverage machine learning algorithms in Bitcoin price prediction, but this paper narrates that the sampling frequency and feature dimensions should also be significantly acknowledged. It first classifies Bitcoin price on the basis of daily price with high dimension and high-frequency price (5-minute interval price) with basic trading features. The outcomes report that Logistic Regression and Linear Discriminant Analysis models perform better for daily data with high-dimensional features, while the XGBoost model outperforms all other machine learning algorithms in the case of high-frequency data. The major significance or implication of this work points to the fact that prudently selected high-dimensional feature sets to compensate for the simplicity of models in Bitcoin's daily price prediction. Therefore, it can be suggested that for daily bitcoin price prediction, precision can be achieved with simplified models if we have multiple features in the data set. Price predicted approaches executed in this paper can be applied to other areas that have similar characteristics like that of Bitcoin.

This study could be potentially extended by employing other machine learning algorithms like Long-short term memory (LSTM), Recurrent Neural Networks (RNN), and so on. Furthermore, additional features like media hype and sentiment on social networks regarding Bitcoins can be considered for an advanced approach to this study.

APPENDIX

Appendix 7.1. Machine Learning Models on Bitcoin Daily Data

Model: Logistic Regression

Accuracy Score: 0.6484018264840182

Table 7.1. Confusion Matrix on Bitcoin Daily Price Using Logistic Regression

	Predicted (-1)	Predicted (1)
Actual (-1)	76	42
Actual (1)	35	66

Table 7.2. Classification Report on Bitcoin Daily Price Using Logistic Regression

	Precision	Recall	F1-Score	Support
-1	0.68	0.64	0.66	118
1	0.61	0.65	0.63	101
Micro Average	0.65	0.65	0.65	219
Macro Average	0.65	0.65	0.65	219
Weighted Average	0.65	0.65	0.65	219

Model: Linear Discriminant Analysis

Accuracy Score: 0.5981735159817352

Table 7.3. Confusion Matrix on Bitcoin Daily Price Using Linear Discriminant Analysis

	Predicted (-1)	Predicted (1)
Actual (-1)	105	13
Actual (1)	75	26

Table 7.4. Classification Report on Bitcoin Daily Price Using Linear Discriminant Analysis

	Precision	Recall	F1-Score	Support
-1	0.58	0.89	0.70	118
1	0.67	0.26	0.37	101
Micro Average	0.60	0.60	0.60	219
Macro Average	0.62	0.57	0.54	219
Weighted Average	0.62	0.60	0.55	219

Model: Random Forest

Accuracy Score: 0.5114155251141552

Table 7.5. Confusion Matrix on Bitcoin Daily Price Using Random Forest

	Predicted (-1)	Predicted (1)
Actual (-1)	71	47
Actual (1)	61	40

Table 7.6. Classification Report on Bitcoin Daily Price Using Random Forest

	Precision	Recall	F1-Score	Support
-1	0.54	0.57	0.56	118
1	0.47	0.45	0.46	101
Micro Average	0.51	0.51	0.51	219
Macro Average	0.51	0.51	0.51	219
Weighted Average	0.51	0.51	0.51	219

Model: XGBoost

Accuracy Score: 0.4748858447488584

Table 7.7. Confusion Matrix on Bitcoin Daily Price Using XGBoost

	Predicted (-1)	Predicted (1)
Actual (-1)	14	104
Actual (1)	11	90

Table 7.8. Classification Report on Bitcoin Daily Price Using XGBoost

	Precision	Recall	F1-Score	Support
-1	0.56	0.12	0.20	118
1	0.46	0.89	0.61	101
Micro Average	0.47	0.47	0.47	219
Macro Average	0.51	0.50	0.40	219
Weighted Average	0.52	0.47	0.39	219

Model: Quadratic Discriminant Analysis

Accuracy Score: 0.4474885844748858

Table 7.9. Confusion Matrix on Bitcoin Daily Price Using Quadratic Discriminant Analysis

	Predicted (-1)	Predicted (1)
Actual (-1)	19	99
Actual (1)	22	79

Table 7.10. Classification Report on Bitcoin Daily Price Using Quadratic Discriminant Analysis

	Precision	Recall	F1-Score	Support
-1	0.46	0.16	0.24	118
1	0.44	0.78	0.57	101
Micro Average	0.45	0.45	0.45	219
Macro Average	0.45	0.47	0.40	219
Weighted Average	0.45	0.45	0.39	219

Model: K-Nearest Neighbors

Accuracy Score: 0.4703196347031963

Table 7.11. Confusion Matrix n Bitcoin Daily Price Using K-Nearest Neighbors

	Predicted (-1)	Predicted (1)
Actual (-1)	11	107
Actual (1)	9	92

Table 7.12. Classification Report on Bitcoin Daily Price Using K-Nearest Neighbors

	Precision	Recall	F1-Score	Support
-1	0.55	0.09	0.16	118
1	0.46	0.91	0.61	101
Micro Average	0.47	0.47	0.47	219
Macro Average	0.51	0.50	0.39	219
Weighted Average	0.51	0.47	0.37	219

Model: Decision Tree

Accuracy Score: 0.5616438356164384

Table 7.13. Confusion Matrix on Bitcoin Daily Price Using Decision Tree

	Predicted (-1)	Predicted (1)
Actual (-1)	64	54
Actual (1)	44	57

Table 7.14. Classification Report on Bitcoin Daily Price Using Decision Tree

	Precision	Recall	F1-Score	Support
-1	0.58	0.68	0.63	118
1	0.53	0.43	0.47	101
Micro Average	0.56	0.56	0.56	219
Macro Average	0.56	0.55	0.55	219
Weighted Average	0.56	0.56	0.55	219

Model: Support Vector Machine

Accuracy Score: 0.4611872146118721

Table 7.15. Confusion Matrix on Bitcoin Daily Price Using Support Vector Machine

	Predicted (1)
Actual (-1)	118
Actual (1)	101

Table 7.16. Classification Report on Bitcoin Daily Price Using Support Vector Machine

	Precision	Recall	F1-Score	Support
-1	0.00	0.00	0.00	118
1	0.46	1.00	0.63	101
Micro Average	0.46	0.46	0.46	219
Macro Average	0.23	0.50	0.32	219
Weighted Average	0.21	0.46	0.55	219

Appendix 2. Machine Learning Models On Bitcoin 5-Minutes Interval Data

Model: Logistic Regression

Accuracy Score: 0.5939269171384457

Table 7.17. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using Logistic Regression

	Predicted (-1)	Predicted (1)
Actual (-1)	4428	87
Actual (1)	3069	118

Table 7.18. Classification Report on Bitcoin 5-Minutes Interval Price Using Logistic Regression

	Precision	Recall	F1-Score	Support
-1	0.59	0.98	0.74	4515
1	0.68	0.06	0.11	3257
Micro Average	0.59	0.59	0.59	7772
Macro Average	0.64	0.52	0.42	7772
Weighted Average	0.63	0.59	0.47	7772

Model: Linear Discriminant Analysis

Accuracy Score: 0.593798250128667

Table 7.19. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using Linear Discriminant Analysis

	Predicted (-1)	Predicted (1)
Actual (-1)	4450	65
Actual (1)	3092	165

Table 7.20. Classification Report on Bitcoin 5-Minutes Interval Price Using Linear Discriminant Analysis

	Precision	Recall	F1-Score	Support
-1	0.59	0.99	0.74	4515
1	0.72	0.05	0.09	3257
Micro Average	0.59	0.59	0.59	7772
Macro Average	0.65	0.52	0.42	7772
Weighted Average	0.64	0.59	0.47	7772

Model: Random Forest

Accuracy Score: 0.5432321152856407

Table 7.21. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using Random Forest

	Predicted (-1)	Predicted (1)
Actual (-1)	2785	1730
Actual (1)	1841	1416

Table 7.22. Classification Report on Bitcoin 5-Minutes Interval Price Using Random Forest

	Precision	Recall	F1-Score	Support
-1	0.61	0.61	0.61	4515
1	0.45	0.44	0.45	3257
Micro Average	0.54	0.54	0.54	7772
Macro Average	0.53	0.53	0.53	7772
Weighted Average	0.54	0.54	0.54	7772

Model: XGBoost

Accuracy Score: 0.5941842511580031

Table 7.23. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using XGBoost

	Predicted (-1)	Predicted (1)
Actual (-1)	4103	412
Actual (1)	2742	515

Table 7.24. Classification Report on Bitcoin 5-Minutes Interval Price Using XGBoost

	Precision	Recall	F1-Score	Support
-1	0.60	0.91	0.72	4515
1	0.56	0.16	0.25	3257
Micro Average	0.59	0.59	0.59	7772
Macro Average	0.58	0.53	0.48	7772
Weighted Average	0.58	0.59	0.52	7772

Model: Quadratic Discriminant Analysis

Accuracy Score: 0.5110653628409676

Table 7.25. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using Quadratic Discriminant Analysis

	Predicted (-1)	Predicted (1)
Actual (-1)	2488	2067
Actual (1)	1733	1524

Table 7.26. Classification Report on Bitcoin 5-Minutes Interval Price Using Quadratic Discriminant Analysis

	Precision	Recall	F1-Score	Support
-1	0.59	0.54	0.56	4515
1	0.42	0.47	0.45	3257
Micro Average	0.51	0.51	0.51	7772
Macro Average	0.50	0.51	0.50	7772
Weighted Average	0.52	0.51	0.51	7772

Model: K-Nearest Neighbors

Accuracy Score: 0.5400154400411734

Table 7.27. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using K-Nearest Neighbors

	Predicted (-1)	Predicted (1)
Actual (-1)	3027	1488
Actual (1)	2087	1170

Table 7.28. Classification Report on Bitcoin 5-Minutes Interval Price Using K-Nea rest Neighbors

	Precision	Recall	F1-Score	Support
-1	0.59	0.67	0.63	4515
1	0.44	0.36	0.40	3257
Micro Average	0.54	0.54	0.54	7772
Macro Average	0.52	0.51	0.51	7772
Weighted Average	0.53	0.54	0.53	7772

Model: Decision Tree

Accuracy Score: 0.5317807514153371

Table 7.29. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using Decision Tree

	Predicted (-1)	Predicted (1)
Actual (-1)	2795	1720
Actual (1)	1923	1334

Table 7.30. Classification Report on Bitcoin 5-Minutes Interval Price Using Decision Tree

	Precision	Recall	F1-Score	Support
-1	0.59	0.62	0.60	4515
1	0.44	0.41	0.43	3257
Micro Average	0.53	0.53	0.53	7772
Macro Average	0.52	0.52	0.52	7772
Weighted Average	0.53	0.53	0.53	7772

Model: Support Vector Machine

Accuracy Score: 0.5654915079773546

Table 7.31. Confusion Matrix on Bitcoin 5-Minutes Interval Price Using Support Vector Machine

	Predicted (-1)	Predicted (1)
Actual (-1)	4155	360
Actual (1)	35	240

Table 7.32. Classification Report on Bitcoin 5-Minutes Interval Price Using Support Vector Machine

	Precision	Recall	F1-Score	Support
-1	0.58	0.92	0.71	4515
1	0.40	0.07	0.12	3257
Micro Average	0.57	0.57	0.57	7772
Macro Average	0.49	0.50	0.42	7772
Weighted Average	0.50	0.57	0.47	7772

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